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Artificial Intelligence in Medicine

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About the journal

Aims and scope

Artificial Intelligence in Medicine publishes original articles from a wide variety of interdisciplinary perspectives concerning the theory and practice of **artificial intelligence (AI)** in **medicine**, **medically-oriented human biology**, and **health care**.

Artificial intelligence in medicine may be characterized as the scientific discipline pertaining to research studies, projects, and applications that aim at supporting decision-based medical tasks through knowledge- and/or data-intensive computer-based solutions that ultimately support and improve the performance of a human care provider.

Artificial Intelligence in Medicine considers for publication manuscripts that have both:

- **Potential high impact in some medical or healthcare domain;**
- **Strong novelty of method and theory related to AI and computer science techniques.**

Artificial Intelligence in Medicine papers must refer to real-world medical domains, considered and discussed at the proper depth, from both the technical and the medical points of view. **The inclusion of a clinical assessment** of the usefulness and potential impact of the submitted work **is strongly recommended**.

Artificial Intelligence in Medicine is looking for novelty in the methodological and/or theoretical content of submitted papers. Such kind of novelty has to be mainly acknowledged in the area of AI and Computer Science. **Methodological papers** deal with the proposal of some strategy and related methods to solve some scientific issues in specific domains. They must show, usually through an experimental evaluation, how the proposed methodology can be applied to medicine, medically-oriented human biology, and health care, respectively. They have also to provide a comparison with other proposals, and explicitly discuss elements of novelty. **Theoretical papers** focus on more fundamental, general and formal topics of AI and must show the novel expected effects of the proposed solution in some medical or healthcare field.

Following the information explosion brought by the diffusion of Internet, social networks, cloud computing, and big-data platforms, Artificial Intelligence in Medicine has broadened its perspective. Particular attention is given to novel research work pertaining to:

- AI-based clinical decision making;
- Medical knowledge engineering;
- Knowledge-based and agent-based systems;
- Computational intelligence in bio- and clinical medicine;
- Intelligent and process-aware information systems in healthcare and medicine;
- Natural language processing in medicine;

- Data analytics and mining for biomedical decision support;
- New computational platforms and models for biomedicine;
- Intelligent exploitation of heterogeneous data sources aimed at supporting decision-based and data-intensive clinical tasks;
- Intelligent devices and instruments;
- Automated reasoning and meta-reasoning in medicine;
- Machine learning in medicine, medically-oriented human biology, and healthcare;
- AI and data science in medicine, medically-oriented human biology, and healthcare;
- AI-based modeling and management of healthcare pathways and clinical guidelines;
- Models and systems for AI-based population health;
- AI in medical and healthcare education;
- Methodological, philosophical, ethical, and social issues of AI in healthcare, medically-oriented human biology, and medicine.

If you are considering submitting to *Artificial Intelligence in Medicine*, make sure that your paper meets the quality requirements mentioned above. English exposition must also be clear and revised with due care. Authors are kindly requested to revise their manuscripts with the help of co-authors that are fluent in English or language editing services before submitting their contribution. Papers written in poor English are likely to be rejected.

The mere application of well-known or already published algorithms and techniques to medical data is not regarded as original research work of interest for *Artificial Intelligence in Medicine*, but it may be suitable for other venues.

Artificial Intelligence in Medicine features the following kinds of papers:

- **Original research contributions:** Theoretical and/or methodological papers about novel approaches;
- **Methodological reviews/surveys:** Papers that collect, classify, describe, and critically analyze research designs, methods and procedures;
- **Position papers:** Papers that gather, describe, and analyze the scientific challenges of a specific field, founding them on the related literature;
- **Editorials:** Editors will occasionally publish editorials;
- **Guest editorials:** Editors can invite guest editors of special issues to publish editorials. Unsolicited editorials will not be considered;
- **Letters to the editor:** Letters from readers shortly discussing and commenting on a topic of interest, for example based on recently published articles in the journal *Artificial Intelligence in Medicine*;
- **Book reviews:** A critical review of recently published books;
- **Erratum:** Some specific corrections to results previously published in the journal *Artificial Intelligence in Medicine*;
- **Historical perspectives:** Papers that describe and critically review some specific aspects in the history of scientific contributions and applications;
- **In memoriam:** Papers describing the life and the main scientific contributions of scientists passed away, having had an important role in the area of artificial intelligence in medicine;
- **PhD projects:** Early publications about more recent research trends, having the goal of allowing PhD candidates to explain their PhD research project and to share it with other scientists interested in the topic. Such type of papers should focus on the overall goals and approaches of PhD research projects, without considering in detail the specific scientific results obtained, which would be the focus of other research articles.

Special Issues are regularly published and included among regular issues. *Artificial Intelligence in Medicine* special issues deal with current theoretical/methodological research or convincing applications related to AI in medicine. Special Issues are managed by one or more guest editors who are outstanding experts on the selected topic. Special Issues of *Artificial Intelligence in Medicine* are directly proposed to potential guest editors by the Editor in Chief, also according to suggestions from the editorial board members. "External" proposals of Special Issues will no longer be considered.

Artificial Intelligence in Medicine does not publish conference volumes or conference papers. However, selected and high-quality research results presented earlier at conferences may be published in *Artificial Intelligence in Medicine*, in the form of a thoroughly revised (rephrased) and extended (including new research results) original research paper.

Information for authors and further details about the editorial process can be found in the *Guide for Authors* section of the *Artificial Intelligence in Medicine* web page.

Journal announcements

Ethics statement

The journal requires authors to confirm at submission if their research needs an Ethics statement. Such statement will confirm that all procedures were performed in compliance with relevant laws and institutional guidelines and have been approved by the appropriate institutional committee(s). This statement should contain the date and reference number of the ethical approval(s) obtained. Authors should also include in the Ethics statement that informed consent was obtained for experimentation with human subjects. The privacy rights of human subjects must always be observed. In case their research does not need an Ethics statement they must state why.

Peer review

This journal follows a single anonymized review process. Your submission will initially be assessed by our editors to determine suitability for publication in this journal. If your submission is deemed suitable, it will typically be sent to a minimum of two reviewers for an independent expert assessment of the scientific quality. The decision as to whether your article is accepted or rejected will be made by our editors.

Read more about [peer review](#).

Our editors are not involved in making decisions about papers which:

- they have written themselves.
- have been written by family members or colleagues.
- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures and peer review will be handled independently of the editor involved and their research group. Read more about [editor duties](#).

Authors may submit a formal appeal request to the editorial decision, provided it meets all the requirements and follows the procedure outlined in [Elsevier's Appeal Policy](#). Only one appeal per submission will be considered and the appeal decision will be final.

Special issues and article collections

The peer review process for special issues and article collections follows the same process as outlined above for regular submissions, except, a guest editor may send the submissions out to the reviewers and may recommend a decision to the journal editor. The journal editor oversees the peer review process of all special issues and article collections to ensure the high standards of publishing ethics and responsiveness are respected and is responsible for the final decision regarding acceptance or rejection of articles.

Open access

We refer you to our [open access information page](#) to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in [Elsevier's Publishing Ethics Policy](#).

Submission declaration

When authors submit an article to an Elsevier journal it is implied that:

- the work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report. See our policy on [multiple, redundant or concurrent publication](#).
- the article is not under consideration for publication elsewhere.
- the article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- if accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

To verify compliance with our journal publishing policies, we may check your manuscript with our screening tools.

Authorship

All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Authors should appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary contact between the journal and the co-authors. Only the

corresponding author's affiliation will be used to determine eligibility for a publishing agreement, and possible discounts related to it. Affiliations of other co-authors are not relevant for eligibility.

Changes to authorship

The editors of this journal generally will not consider changes to authorship once a manuscript has been submitted. It is important that authors carefully consider the authorship list and order of authors and provide a definitive author list at submission.

The policy of this journal around authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes additions, deletion, or rearrangement of author names.
- Requests to change authorship must be made by the corresponding author using the [Authorship Change Request form](#). The corresponding author must provide the reason for the request to the journal editor with written confirmation from all authors, including any authors being added or removed, that they agree with the changes. Requests which do not comply with the instructions outlined in the form will not be considered.
- This journal does not allow authorship changes after acceptance. This includes additions, deletions, or the rearrangement of author names, including changes to the corresponding author.
- The review process may be paused while a change in authorship request is being considered.
- Change requests made by the authors without the [Authorship Change Request form](#) or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

- Employment
- Consultancies
- Stock ownership
- Honoraria
- Paid expert testimony
- Patent applications or registrations
- Grants or any other funding
- Affiliation with the journal as an Editor or Advisory Board Member

The [declarations tool](#) should always be completed.

Authors with a journal affiliation to declare should enter the following text under “Other Activities” within the [declarations tool](#) and should inform the journal and publisher prior to completing the submission process:

Given their role as [AUTHOR JOURNAL ROLE TITLE], [AUTHOR NAME] had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Editorial disclosure statements will be included as a footnote and/or in the declaration of competing interest section of the article.

Authors with no competing interests to declare should select the option "I have nothing to declare".

The resulting Word document containing your declaration should be uploaded at the "attach/upload files" step in the submission process. It is important that the Word document is saved in the .doc/.docx file format. Author signatures are not required.

Funding sources

Authors must disclose any funding sources who provided financial support for the conduct of the research and/or preparation of the article. The role of sponsors, if any, should be declared in relation to the study design, collection, analysis and interpretation of data, writing of the report and decision to submit the article for publication. If funding sources had no such involvement this should be stated in your submission.

List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions on the program or type of grants, scholarships and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, it is recommended to include the following sentence:

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI use

Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

Generative AI and AI-assisted technologies ("AI tools") have been rapidly adopted in research and scholarly publishing, offering significant benefits while requiring clear standards for responsible and transparent use.

Authors preparing a manuscript for this journal may use AI tools to support them. However, these tools must never be used as a substitute for human critical thinking, expertise and evaluation. AI tools may only be applied with human oversight and control.

Please read Elsevier's detailed author policy on the use of AI tools, as stated in [Elsevier's generative AI policies for journals](#).

Ultimately, authors are responsible and accountable for the contents of their work. This includes accountability for:

- Carefully reviewing and verifying the accuracy, comprehensiveness, and impartiality of all AI-generated output (including checking the sources, as AI-generated references can be incorrect or fabricated).
- Editing and adapting all material thoroughly to ensure the manuscript represents the author's authentic and original contribution and reflects their own analysis, interpretation, insights and ideas.
- Ensuring accuracy and originality of any AI generated images and provide appropriate attribution where relevant.
- Ensuring the use of any tools or sources, AI-based or otherwise, is made clear and transparent to readers. If AI tools have been used, we require a disclosure statement upon submission; please see example below.
- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the references list. An example:

Title of new section: *Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.*

Statement: *During the preparation of this work the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

The declaration does not apply to the use of basic tools, such as tools used to check grammar, spelling and references. In addition, authors are not required to disclose the use of AI tools or features within specialist disability-related assistive technology, provided these are used solely for accessibility purposes.

If you have nothing to disclose, you do not need to add a statement.

Please note: to protect authors' rights and the confidentiality of their research, this journal does not currently allow academic editors and reviewers to upload an unpublished manuscript into a generative AI tool, as is stated in [Elsevier's generative AI policies for journals](#). However, AI tools may be used in a supportive capacity by editors and reviewers to improve structure and language of the review report or editorial decision letter or to conduct a background literature search. Elsevier is actively evaluating compliant AI tools and may revise this policy in the future.

Preprints

Preprint sharing

Authors may share preprints in line with Elsevier's [article sharing policy](#). Sharing preprints, such as on a preprint server, will not count as prior publication.

We advise you to read our policy on [multiple, redundant or concurrent publication](#).

In support of [Open Science](#), this journal offers its authors a free preprint posting service on [SSRN](#), Elsevier's preprint and early research repository.

During submission to this journal, you can choose to post your manuscript on SSRN, and it will be made publicly available as soon as it passes the journal's initial desk review.

As a preprint on SSRN, your manuscript will benefit from:

- Early registration with a preprint DOI (Digital Object Identifier)
- A link from the preprint to the version of record if published in this Elsevier journal

- Preprint posting, sharing, and download availability that facilitates collaboration and early citations

Your decision to post or not post your preprint will have no effect on the editorial process or publication outcome with the journal. For additional information, please consult the [SSRN Terms of Use](#) and [FAQs](#).

It is expected that the corresponding author will seek approval from all co-authors before agreeing to post the manuscript publicly, prior to peer review, on SSRN.

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist](#) (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth")

and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).

- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

Jurisdictional claims

Elsevier respects the decisions taken by its authors as to how they choose to designate territories and identify their affiliations in their published content. Elsevier's policy is to take a neutral position with respect to territorial disputes or jurisdictional claims, including, but not limited to, maps and institutional affiliations. For journals that Elsevier publishes on behalf of a third party owner, the owner may set its own policy on these issues.

- Maps: Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries*". During the review process, Elsevier's editors may request authors to change maps if these guidelines are not followed.
- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

Studies in humans

Authors must follow [ethical guidelines](#) for research involving humans, their data, and/or their biological materials (e.g. organs, tissues). Authors should ensure that they adhere to the following:

- All studies must be performed in compliance with relevant laws, regulations, and institutional guidelines and that the appropriate institutional committee(s) has/have approved them. For instance, medical research involving humans should be carried out in accordance with the World Medical Association [Declaration of Helsinki: Ethical principles for medical research involving human participants](#) and should follow the International Committee of Medical Journal Editors (ICMJE) [recommendations for the conduct, reporting, editing and publication of scholarly work in medical journals](#) and aim to be representative of human populations in terms of sex, age and ethnicity. Sex and gender terms should be used correctly, as outlined in the “Reporting Sex- and Gender-based analyses” section of this journal’s Guide for Authors.
- Informed consent must be obtained for research with human participants, including their data or biological materials. Consent must be given without coercion, and the privacy rights of human participants must always be observed. If your study contains human participants, please read Elsevier’s full [human research policy](#) for further information.
 - Authors should be aware that written (or electronic) consent is required for certain studies under national laws. In studies where only verbal informed consent was obtained, authors should include justification for the absence of written consent, the name of the approving ethics committee, and a statement on how verbal consent was recorded.
 - Consent for publication should also be obtained.
- Authors must obtain and retain explicit written consent for any identifiable participant materials (e.g. case details, videos, recordings, images, photographs, and illustrations). All participant information must be de-identified: do not include names or personal identifiers (e.g., initials, ID numbers, DOB)
- Images should be used only when scientifically necessary and with explicit consent, and nonessential identifying details must be removed. If identifiers are altered to ensure anonymity, authors must confirm that scientific meaning is not affected. Fully anonymized images (e.g., x-rays, scans) that contain no identifying marks and are not accompanied by identifying text do not require formal consent, though informing the patient is good practice. Case reports always require consent. Simply obscuring identity (e.g., eye bars or blurring) is not sufficient without consent.
- Written consents must be retained by the authors. They should not be provided to this journal unless this is specifically requested in exceptional circumstances, for example, when a legal issue arises. Only then should you provide copies of the consents, or evidence that all relevant consents

were obtained.

- Appropriate consents and safeguards should be in place for research with vulnerable populations, as described in Elsevier's [human research policy](#). In many countries, additional ethical approval may be needed for research with those who lack the capacity to consent (e.g. children or those with mental health or neurological conditions including dementia).
- For the full policy, see [“Consent requirements for case reports, personal information, and images.”](#)

All studies involving human participants require a statement that details any ethical approval(s), guidelines followed, confirmation of informed consent, and any other necessary information in line with Elsevier's [human research policy](#). The statement should contain the date and reference number of any ethical approval(s) obtained

If ethical approval is not required by national laws or where a study has been granted an exemption from an ethics committee, the statement must explain why the study is exempt and confirm that the appropriate protocols for protecting the rights and privacy of all participants were utilized during the execution of the research (e.g. no coercion to participate, full disclosure of study requirements and risks, consent of participants (or their Legal Representatives), no release of participant data without their knowledge, etc.).

Human biological material

For all studies that use human organs or tissues, sufficient evidence must be provided that these were procured in line with [WHO Guiding Principles on Human Cell, Tissue and Organ Transplantation](#). The source of the organs or tissues used in clinical research must be transparent and traceable and must not be sourced from executed prisoners or prisoners of conscience, consistent with recommendations by [Global Rights Compliance on Mitigating Human Rights Risks in Transplantation Medicine](#).

For clinical studies, a statement of informed consent having been obtained from a patient or their legal representative, paired with ethical approval for the study from a suitable institution, as required by the policies of the journal, may be considered sufficient evidence, but the journal reserves the right to request additional evidence in cases where it feels this is not sufficient.

If the manuscript describes organ transplantation authors must additionally declare within the manuscript that:

- Autonomous consent, free from coercion, was obtained from the donor(s) or their next of kin.
- Organs and/or tissues were not sourced from executed prisoners or prisoners of conscience.

Clinical Trials

This journal follows the [International Committee of Medical Journal Editors \(ICMJE\) clinical trials registration guidelines](#).

Registration

Clinical trials must be registered in a public trials registry. Purely observational studies, in which the assignment of the medical intervention is not at the discretion of the investigator, do not require registration.

Some important points:

- Trials must be registered at or before the onset of patient enrolment.
- The clinical trial registration number must be included in the manuscript, preferably at the end of the article abstract.
- A clinical trial is defined as any research study that prospectively assigns human participants, or groups of humans, to one or more health-related interventions to evaluate the effects of health outcomes.
- Health-related interventions include any intervention used to modify a biomedical or health-related outcome such as drugs, surgical procedures, devices, behavioral treatments, dietary interventions, and process-of-care changes.
- Health outcomes include any biomedical or health-related measures obtained in patients or participants, including pharmacokinetic measures and adverse events.

Results

Authors are required to disclose all posting in registries of results of the same or closely related work. Editors will not consider results to be a prior publication if they have already been posted in the same clinical trials registry in which primary registration resides, as long as the results are presented in the form of a brief structured abstract (fewer than 500 words) or table.

Disclosing results in other circumstances, such as in an investors' meeting, for example, is discouraged and may jeopardize consideration of your manuscript by this journal.

Reporting

Authors are encouraged to follow CONSORT guidelines when presenting randomized controlled trials, and provide the CONSORT checklist at manuscript submission, with an accompanying flow diagram illustrating the progress of patients through the trial - including recruitment, enrolment, randomization, withdrawal, completion and a description of the randomization procedure.

- Read the [CONSORT guidelines](#).
- Follow the [CONSORT checklist](#).

Delayed publication

If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our [Journal Article Publishing Support Center](#) at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

Writing and formatting

File format

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.

- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.
- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.
- Use spell-check and grammar-check functions to avoid errors.

We advise you to read our [Step-by-step guide to publishing with Elsevier](#).

LaTeX

We encourage you use our [LaTeX template](#) when preparing a LaTeX submission. You will be asked to provide all relevant editable source files upon submission or revision.

Support for your LaTeX submission:

- [LaTeX submission instructions and templates](#)
- Journal Article Publishing Support Center [LaTeX FAQs and support](#)
- Researcher Academy's [Beginners' guide to writing a manuscript in LaTeX](#)

Title page

You are required to include the following details in the title page information:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.
- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. If needed, you can add your name between parentheses in your own script after the English transliteration.

- **Affiliations.** Add affiliation addresses, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full postal address of each affiliation, including the country name and, if available, the email address of each author.
- **Corresponding author.** Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, methodology and materials. It is important that the email address and contact details of your corresponding author are kept up to date during the submission and publication process.
- **Present/permanent address.** If an author has moved since the work described in your article was carried out, or the author was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should briefly state the purpose of your research, principal results and major conclusions. Some guidelines:

- Abstracts must be able to stand alone as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are encouraged to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example [article highlights](#) and read about the benefits of their inclusion.

Units, classifications codes and nomenclature

Units

This journal requires you to use the international system of units (SI) which follows internationally accepted rules and conventions. If other units are mentioned within your article, you should provide the equivalent unit in SI.

Math formulae

- Submit math equations as editable text, not as images.
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Tables must be submitted as editable text, not as images. Some guidelines:

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- Place any table notes below the table body.
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[3] Strunk Jr W, White EB. The elements of style. 4th ed. New York: Longman; 2000.

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[4] Mettam GR, Adams LB. How to prepare an electronic version of your article. In: Jones BS, Smith RZ, editors. Introduction to the electronic age, New York: E-Publishing Inc; 2023, p. 281-304.

Reference to a website:

[5] Cancer Research UK. Cancer statistics reports for the UK, <http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/>; 2023 [accessed 13 March 2023].

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[6] Oguro M, Imahiro S, Saito S, Nakashizuka T. Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset]. Mendeley Data. 2015. <https://doi.org/10.17632/xwj98nb39r.1>.

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[7] Coon E, Berndt M, Jan A, Svyatsky D, Atchley A, Kikinzon E, et al. Advanced Terrestrial Simulator (ATS). Version 0.88 [software]. Zenodo; 2020 Mar 25. <https://doi.org/10.5281/zenodo.3727209>.

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